

Sea Level Rise and Urban Adaptation

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The seas are rising

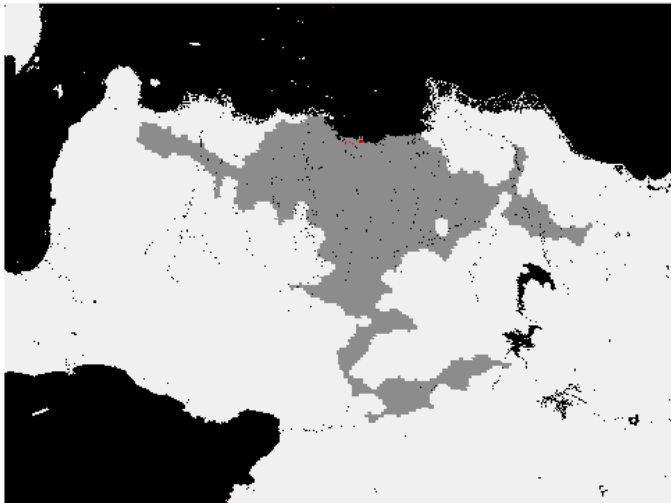
- Global mean sea level rise
 - 0.40-0.69 m for 2°C warming by 2100 (Depsky et al. 2023)
 - Possibly much more via land subsidence, ice-sheet instability
- Adaptation will be shaped by markets, politics, and institutions

Asian cities are highly exposed

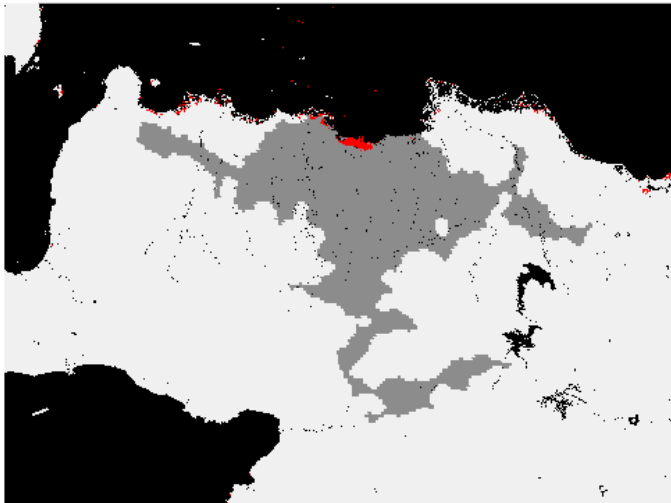
Rank	SLR of 1 m		SLR of 2 m		SLR of 3 m	
	City	%	City	%	City	%
1	Osaka	4.8	Bangkok	55.5	Bangkok	94.4
2	Jakarta	2.2	Shanghai	19.0	Shanghai	80.2
3	Tokyo	2.2	Manila	11.8	Suzhou	31.3
4	Lagos	1.7	Osaka	9.9	Lagos	25.7
5	Mumbai	1.1	Jakarta	9.5	Manila	21.3

Hsiao (2025). Sea Level Rise and Urban Infrastructure. *AEA P&P*.

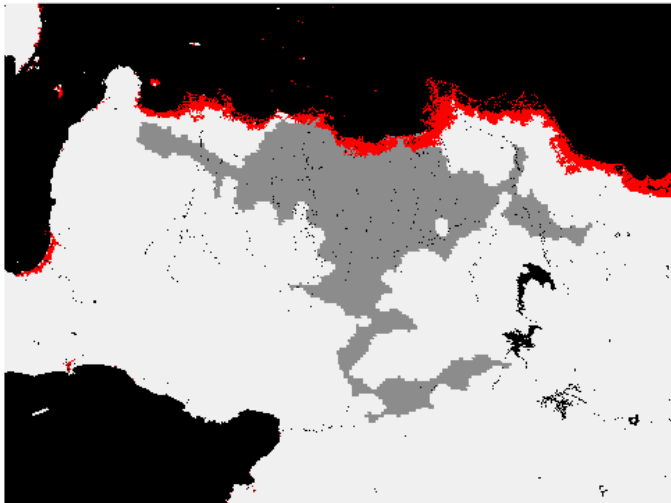
Jakarta is home to 42 million



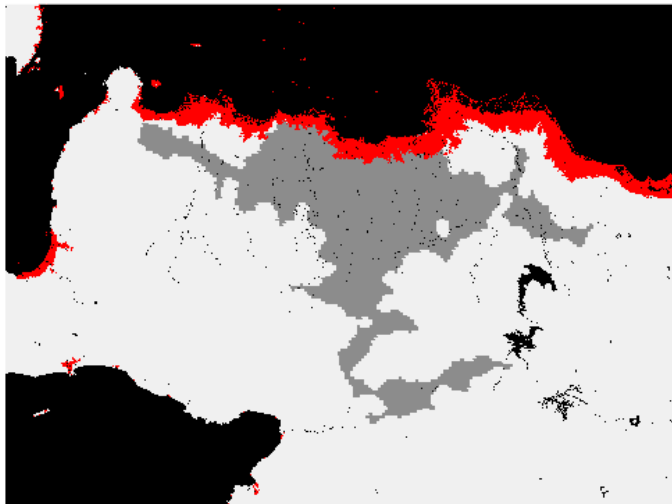
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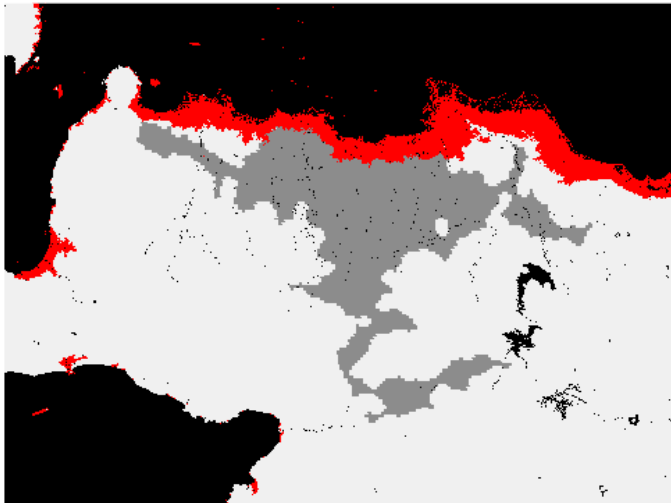
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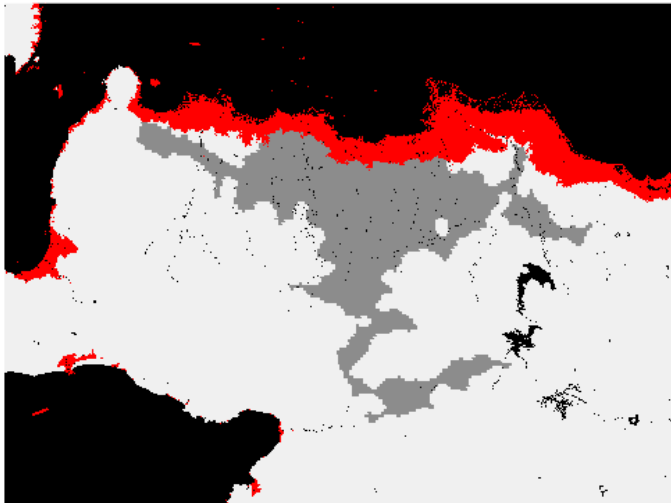
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Adaptation faces political frictions

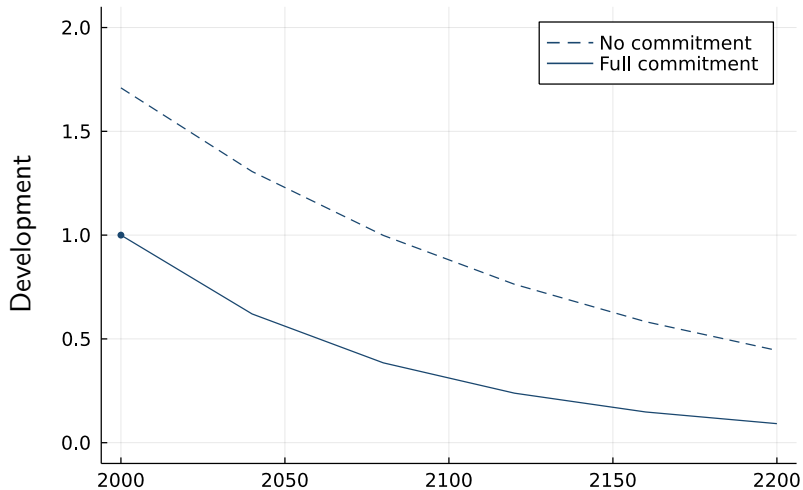
- Sea wall for Jakarta at up to \$60B
 - Evaluated with granular spatial data
- Risk of coastal lock-in
 - Moral hazard as defense responds to development
 - Development anticipates defense at public cost
 - After sunk development, government forced to defend

Hsiao (2025). Sea Level Rise and Urban Adaptation in Jakarta.

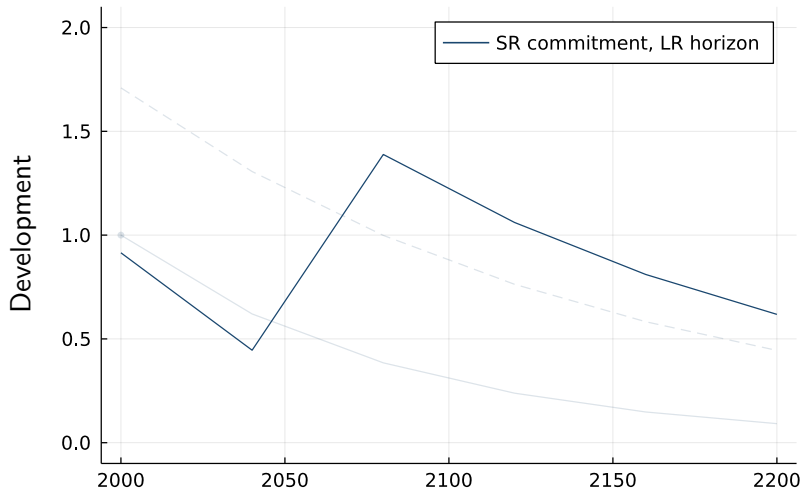
Potential distortions

- Spatial: $d_1^{\text{co}} \uparrow$ induces $g_1^{\text{co}} \uparrow$, $d_1^{\text{in}} \downarrow$
 - Coastal residents can force local government
 - Local government can force national government
- Dynamic: $d_1^{\text{co}} \uparrow$ induces $g_2^{\text{co}} \uparrow$, $d_2^{\text{in}} \downarrow$
 - Current government can force future government
 - Future government cannot regulate current government

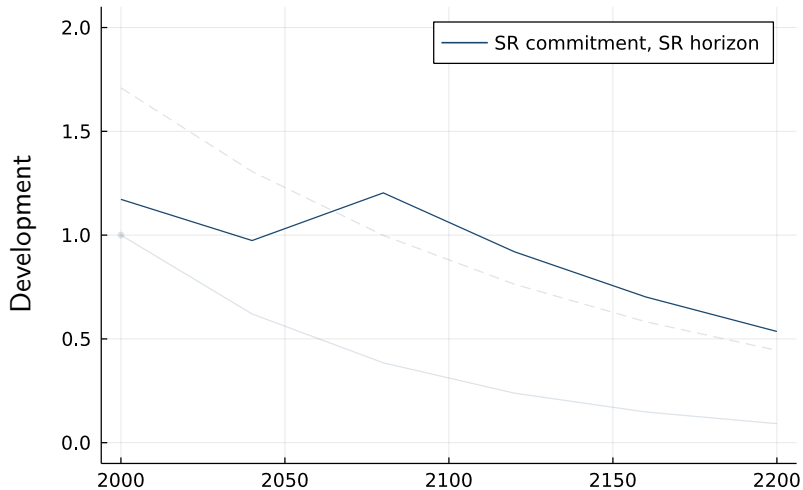
Commitment helps, subject to politics



Commitment helps, subject to politics



Commitment helps, subject to politics



Adaptation can be regressive

- Spatial sorting on flood exposure
 - If the rich are more willing to pay for flood safety
 - Then they move to flood-safe areas, bidding up housing prices
- The rich crowd out the poor in pursuit of higher ground

Hsiao (2024). Sea Level Rise and Urban Inequality. *AEA P&P*.

Sorting drives growing inequality

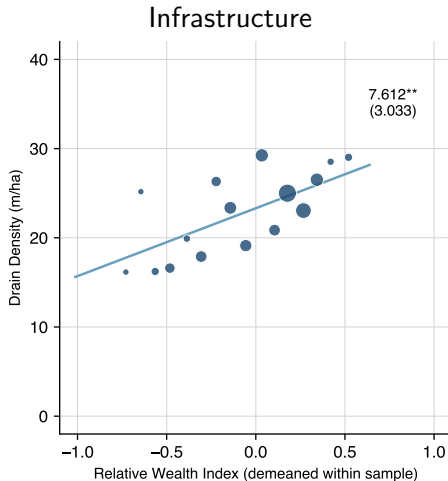
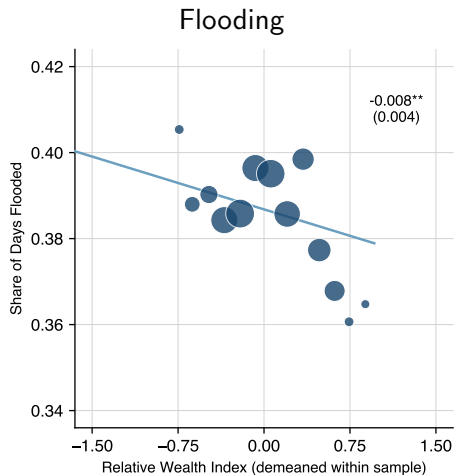
Flooding in Jakarta	Low wages	High wages	L/H
Current	0.88	0.62	1.40
Projected			
1m sea level rise	1.03	0.68	1.52
5m sea level rise	5.77	2.20	2.62
Projected, no sorting			
1m sea level rise	1.02	0.73	1.39
5m sea level rise	5.47	3.72	1.47

With implications for adaptation policy

- Theory and data from Indian cities
 - New satellite-based measures of flooding
 - Customized for urban environments

Hsiao & Patel (2026). Urban Flooding in India.

For the rich, less flooding and more infrastructure



But the poor are compensated

- Two locations $\{H, L\}$, two groups $\{R, P\}$
 - WTP $\{\gamma_R, \gamma_P\}$ to avoid flooding
- (Regressive) sorting happens iff $\gamma_R > \gamma_P$
 - But market clears on aggregate $\bar{\gamma}$
 - For the poor, over-compensation through low prices

Adaptation policy faces a trade-off

- Because modes of adaptation interact
 - Policy: government infrastructure (public, in situ)
 - Response: migration and sorting (private, ex situ)
- Protecting location H with infrastructure
 - Reduces exposure gaps and aggregate losses
 - But leads to gentrification and uniformly expensive housing

Questions

How can cities adapt? How can governments help?